



NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI – 15
DEPARTMENT OF MECHANICAL ENGINEERING

Cycle Test-2

Programme and specialization: B.Tech.- Instrumentation and Control Engineering

Course code/Title : MEIR11/Basics of Mechanical Engineering

Max. Marks: 20

Date: 26.11.2025

Duration: 60 Minutes

Answer all the questions.

Part-A

1. What is the function of the compressor in V.C. Refrigeration cycle? ✓ [1 M]
2. Mention two important thermal properties of a good refrigerant. [1 M]
3. The power required for a refrigerator is 2 kW per ton of refrigeration. Determine the amount of Heat rejected per ton of refrigeration. ✓ [1 M]
4. What is meant by superheated steam and wet steam? ✓ [1 M]
5. Compare closed-cycle and open-cycle GT cycles. [2 M]
6. With the help of T-S chart and schematic diagram explain the Rankine cycle. [4 M]

Part-B

1. Determine the ratio for a spur gear mesh in which the gear has 48 teeth and the pinion has 24 teeth. [1 M]
2. Draw the layout of a four-wheeled electrical vehicle (EV), shows the main components. [2 M]
3. Modern cars brake safely even on slippery surfaces. Identify the braking system responsible for this feature and describe its working principle with a simple block diagram. [3 M]
4. Classify advanced engineering materials and provide one example for each category. [2 M]
5. Define the following mechanical properties:
a)Strength b)Stiffness c)Ductility d) Malleability [2 M]